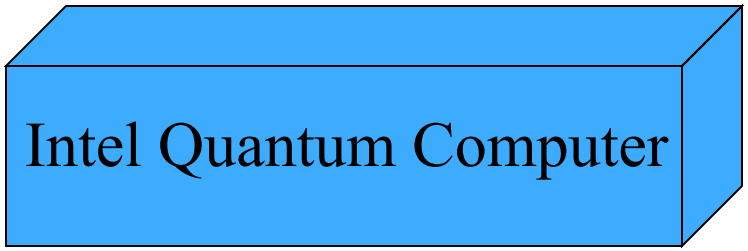
Do Not Read This

Introduction to Quantum Error Correction

Weatherproofing Quantum Computers





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Technology is the Quest for Control

"To be an Error and to be Cast out is Part of [Nature's] Design."
-*William Blake*

Things Fall Apart -*Chinua Achebe*

EVENTUALLY your desktop computer hardware will ERROR.

EVENTUALLY the information on a hard drive will be CORUPTED.

Amount of useful computer cycles before such an error occurs (CURRENTLY) = large large number

—

Exist robust physical systems which serve as classical computers

Quantum Things Fall Apart Fast

Amount of quantum computer cycles before an error ≠ large number (CURRENTLY?)

Quantum Computing is Bunk

Ways Quantum Computers Fail to Quantum Compute

Decoherence

$$(\alpha|0\rangle + \beta|1\rangle) \otimes |\text{environment}\rangle \rightarrow \alpha|0\rangle \otimes |\text{environment}_A\rangle + \beta|1\rangle \otimes |\text{environment}_B\rangle$$

Lack of Unitary Control

attempting to apply unitary evolution **U** instead results in **V**

or (worse) results in non-unitary evolution $\boldsymbol{\rho} \rightarrow \sum_k \mathbf{A}_k \boldsymbol{\rho} \mathbf{A}_k^\dagger$

Measurements are faulty

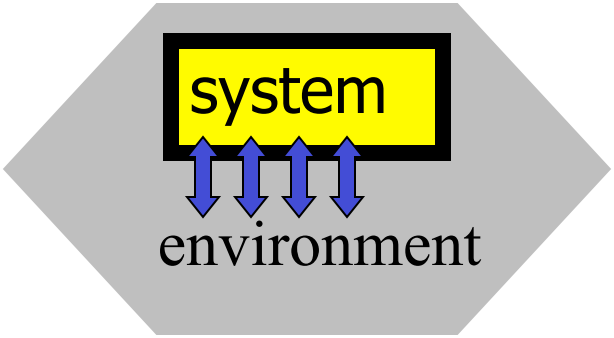
measurement result is noisy, incorrect result obtained

Quantum Computing Disappearing Act

qubits disappear (leakage of computing states)

What is the Problem?

Real quantum systems are open quantum systems!



Full Hilbert space $\mathcal{H} = \mathcal{H}_S \otimes \mathcal{H}_B$

Evolution is $\mathbf{U} = \exp[-i\mathbf{H}_{SB}t]$

Full Hamiltonian:

System evolution — environment evolution — **Interaction causing DECOHERENCE**

$$\mathbf{H}_{SB} = \mathbf{H}_S \otimes \mathbf{I}_B + \mathbf{I}_S \otimes \mathbf{H}_B + \mathbf{H}_I$$

System's pure states become entangled with environment:

$|\psi_0\rangle = |\psi\rangle_S \otimes |\psi\rangle_B \rightarrow \sum_n c_n |n\rangle \otimes |n\rangle \quad \rho_0 = |\psi_0\rangle_S \, {}_S\langle\psi_0| \rightarrow$ mixed state

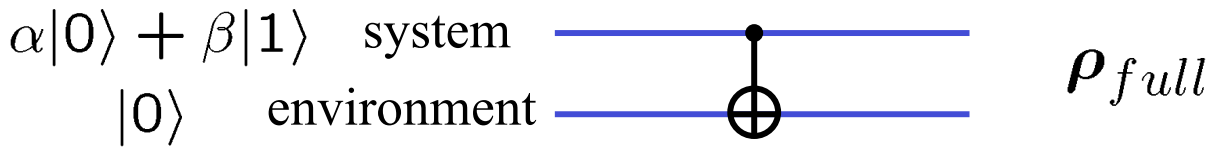
System evolution (by itself) is no longer unitary

Coherent properties of system's quantum information lost

Quantum Error Example

"To err is human, to really foul things up requires a computer." -Anonymous

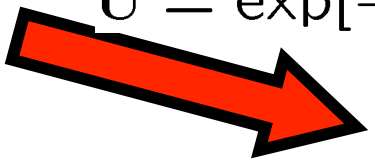
$$\mathbf{U} = \exp[-i\mathbf{H}_{SB}t]$$



$$(\alpha|0\rangle + \beta|1\rangle) \otimes |0\rangle \rightarrow \alpha|00\rangle + \beta|11\rangle$$

$$\boldsymbol{\rho}_{sys} = \mathrm{Tr}_{env}\left[\boldsymbol{\rho}_{full}\right] = \langle 0_{env}|\boldsymbol{\rho}_{full}|0_{env}\rangle + \langle 1_{env}|\boldsymbol{\rho}_{full}|1_{env}\rangle$$

Before $\mathbf{U}$: $\boldsymbol{\rho}_{sys} = \begin{bmatrix} |\alpha|^2 & \alpha\beta^* \\ \alpha^*\beta & |\beta|^2 \end{bmatrix}$



After $\mathbf{U}$: $\boldsymbol{\rho}_{sys} = \begin{bmatrix} |\alpha|^2 & 0 \\ 0 & |\beta|^2 \end{bmatrix}$

Not unitary:

$$\det\left[\mathbf{U}\boldsymbol{\rho}\mathbf{U}^\dagger\right] = \det\left[\mathbf{U}\right]\det\left[\mathbf{U}^\dagger\right]\det\left[\boldsymbol{\rho}\right] = \det\left[\boldsymbol{\rho}\right]$$

Operator Sum Representation

$\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B$



Assume initially $\boldsymbol{\rho} = \boldsymbol{\rho}_A \otimes \boldsymbol{\rho}_B$

$$\boldsymbol{\rho}_{out} = \mathrm{Tr}_B\left[\boldsymbol{\rho}_{AB}\right] = \mathrm{Tr}_B\left[\mathbf{U}\left(\boldsymbol{\rho}_A \otimes \boldsymbol{\rho}_B\right)\mathbf{U}^\dagger\right]$$

Spectrally decompose $\boldsymbol{\rho}_B$: $\boldsymbol{\rho}_B = \sum_\nu p_\nu |\nu\rangle\langle\nu|$

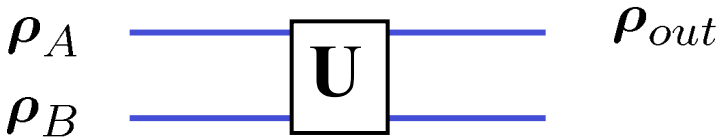
$$\begin{aligned}
\boldsymbol{\rho}_{out} &= \sum_\nu p_\nu \mathrm{Tr}_B\left[\mathbf{U}\left(\boldsymbol{\rho}_A \otimes |\nu\rangle\langle\nu|\right)\mathbf{U}^\dagger\right] \\
&= \sum_{\nu,\mu} p_\nu \langle\mu|\mathbf{U}\left(\boldsymbol{\rho}_A \otimes |\nu\rangle\langle\nu|\right)\mathbf{U}^\dagger|\mu\rangle \\
&= \sum_{\nu\mu} \sqrt{p_\nu}\langle\mu|\mathbf{U}|\nu\rangle \boldsymbol{\rho}_A \langle\nu|\mathbf{U}^\dagger|\mu\rangle\sqrt{p_\nu}
\end{aligned}$$

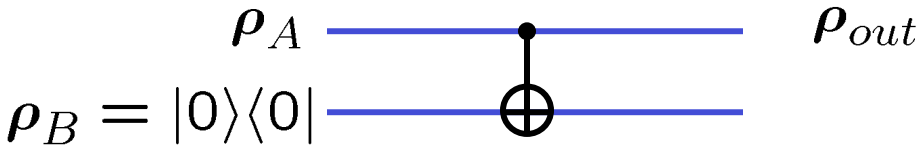
B states

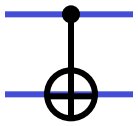
$$\boldsymbol{\rho}_{out} = \sum_k \mathbf{A}_k \boldsymbol{\rho}_A \mathbf{A}_k^\dagger \quad \text{where } \mathbf{A}_{k=(\mu,\nu)} = \sqrt{p_\nu}\langle\mu|\mathbf{U}|\nu\rangle$$

Exercise: Unitarity of $\mathbf{U}$ implies $\sum_k \mathbf{A}_k^\dagger \mathbf{A}_k = \mathbf{I}$

OSR Example







$$= |0\rangle\langle 0| \otimes \mathbf{I} + |1\rangle\langle 1| \otimes \mathbf{X} = \frac{1}{2}(\mathbf{II} + \mathbf{ZI} + \mathbf{IX} - \mathbf{ZX})$$

$$\boldsymbol{\rho}_B = \textstyle\sum_\nu p_\nu |\nu\rangle\langle\nu| \rightarrow \boldsymbol{\rho}_B = |0\rangle\langle 0|$$

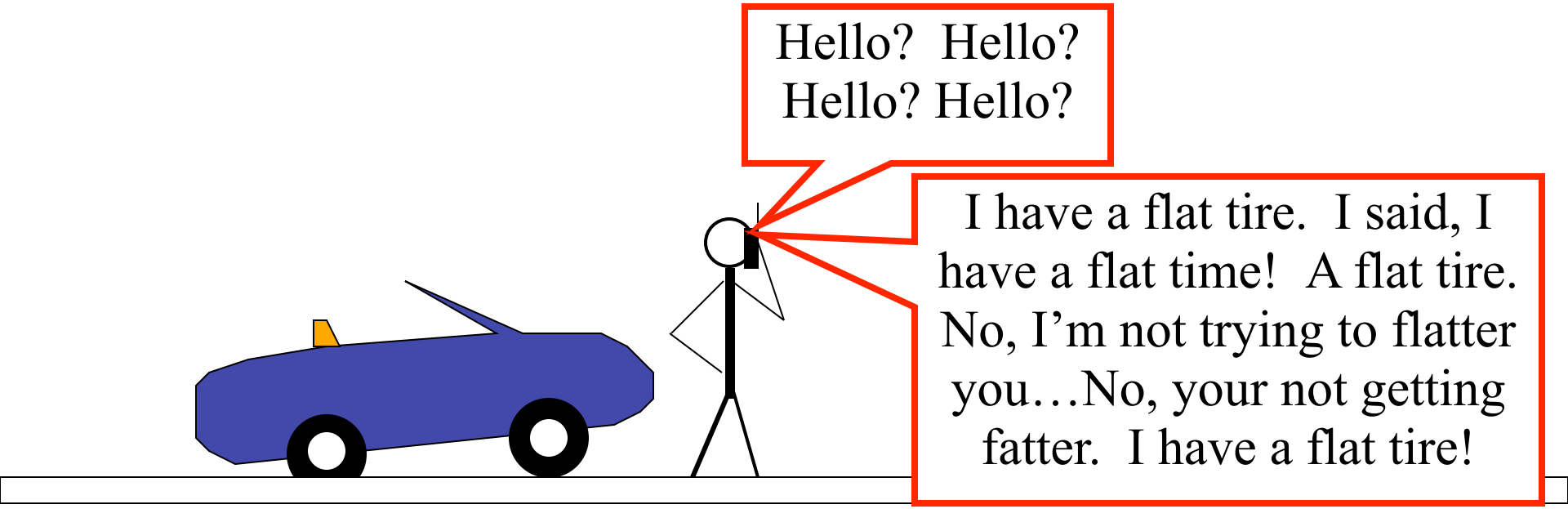
$$\mathbf{A}_{k=(\mu,\nu)} = \sqrt{p_\nu}\langle\mu|\mathbf{U}|\nu\rangle \rightarrow \mathbf{A}_\mu = \langle\mu|\mathbf{U}|0\rangle$$

$$\mathbf{A}_0 = \langle 0|\mathbf{U}|0\rangle = \langle 0|\frac{1}{2}[\mathbf{II} + \mathbf{ZI} + \mathbf{IX} - \mathbf{ZX}]|0\rangle = \frac{1}{2}(\mathbf{I} + \mathbf{Z}) = |0\rangle\langle 0|$$

$$\mathbf{A}_1 = \langle 1|\mathbf{U}|0\rangle = \langle 1|\frac{1}{2}[\mathbf{II} + \mathbf{ZI} + \mathbf{IX} - \mathbf{ZX}]|0\rangle = \frac{1}{2}(\mathbf{I} - \mathbf{Z}) = |1\rangle\langle 1|$$

$$\boldsymbol{\rho}_{out} = \sum_k \mathbf{A}_k \boldsymbol{\rho}_A \mathbf{A}_k^\dagger \rightarrow \boldsymbol{\rho}_{out} = |0\rangle\langle 0|\boldsymbol{\rho}_A|0\rangle\langle 0| + |1\rangle\langle 1|\boldsymbol{\rho}_A|1\rangle\langle 1|$$

Noisy Cell Phone



Communication over a noisy CHANNEL can be overcome via

ENCODING

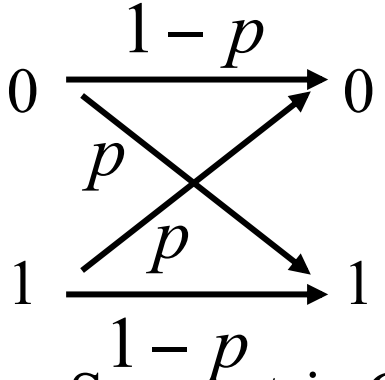
"Hello?" = "Hello? Hello? Hello? Hello?" [using redundancy to encode "Hello"]

Simple Repetition Code

Encode: $0 \rightarrow 000\ldots000$

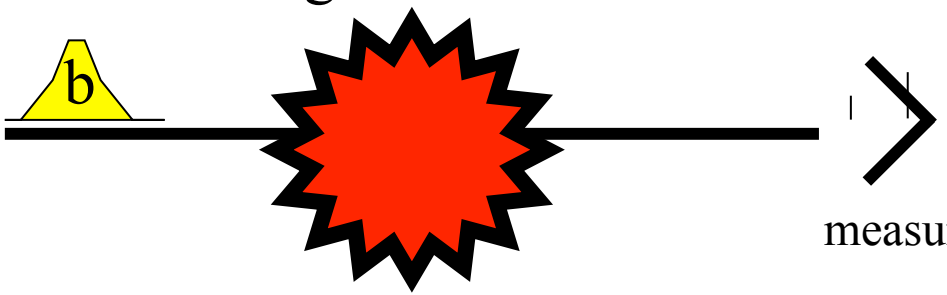
$1 \rightarrow 111\ldots111$

n copies



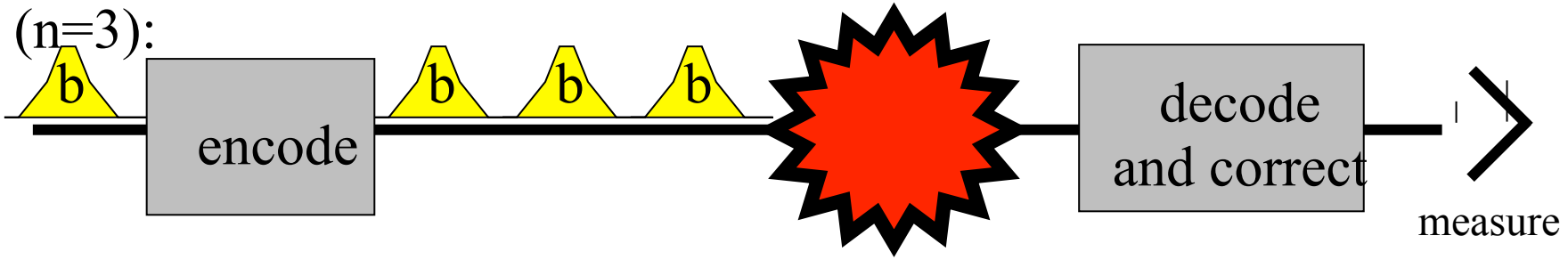
Binary Symmetric Channel

No encoding:



Probability of error = p

Encoding (n=3):



Probability of error

$$= 3p^2(1-p) + p^3 = 3p^2 - 2p^3 < p, p < \tfrac{1}{2}$$

1994 Reasons to be a Pessimist

No cloning:

No cloning of quantum information:

$$\nexists \mathcal{T} || \psi \rangle \rightarrow |\psi\rangle \otimes |\psi\rangle, \forall |\psi\rangle$$

Quantum repetition code impossible:

$$|\psi\rangle \nrightarrow |\psi\rangle \otimes |\psi\rangle \otimes |\psi\rangle$$

Continuous set of errors:

Classical errors on digital information are discrete.

Quantum errors on classical information are continuous.

Continuous set of operations:

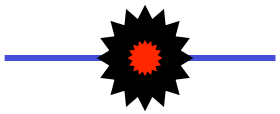
Continuous set of unitary operations: infinite precision needed to implement a gate?

Measurement destroys coherence:

How can one decode without destroying the information?

Baby Steps

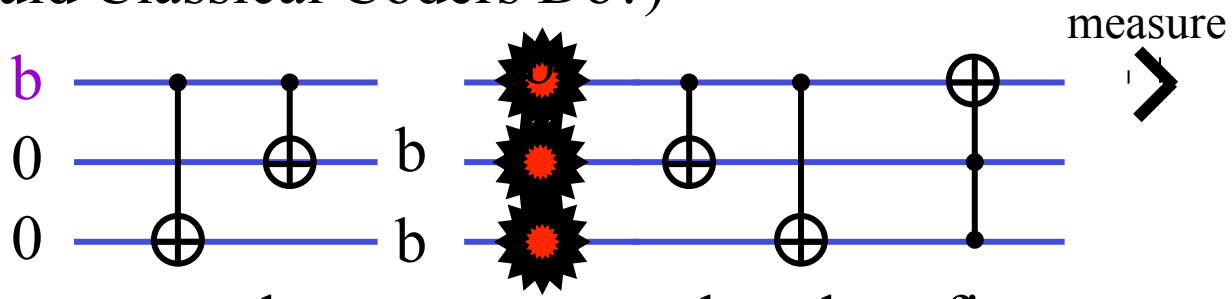
Bit Flip Channel: $\boldsymbol{\rho}_{out} = \sum_k \mathbf{A}_k \boldsymbol{\rho}_{in} \mathbf{A}_k^\dagger = (1-p)\boldsymbol{\rho}_{in} + p\boldsymbol{\sigma}_x \boldsymbol{\rho}_{in} \boldsymbol{\sigma}_x$



$$\mathbf{A}_1 = \sqrt{1-p}\mathbf{I} = \begin{bmatrix} \sqrt{1-p} & 0 \\ 0 & \sqrt{1-p} \end{bmatrix}, \mathbf{A}_2 = \sqrt{p}\boldsymbol{\sigma}_x = \begin{bmatrix} 0 & \sqrt{p} \\ \sqrt{p} & 0 \end{bmatrix}$$

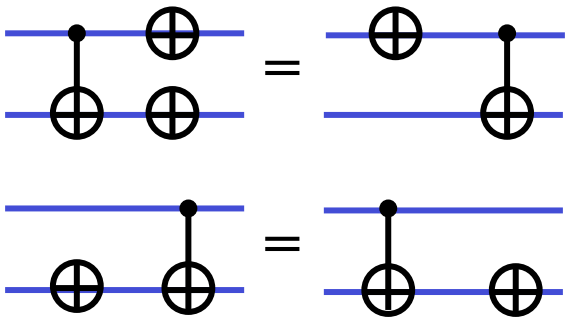
WWCCD? (What Would Classical Coders Do?)

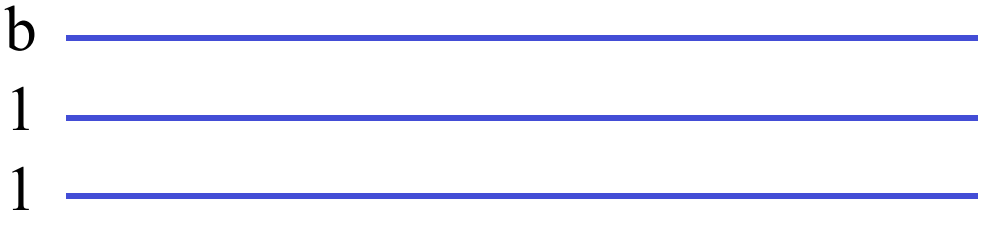
0 □000, 1 □111



100 □ 111 □ 101 □110 □ 110

identities



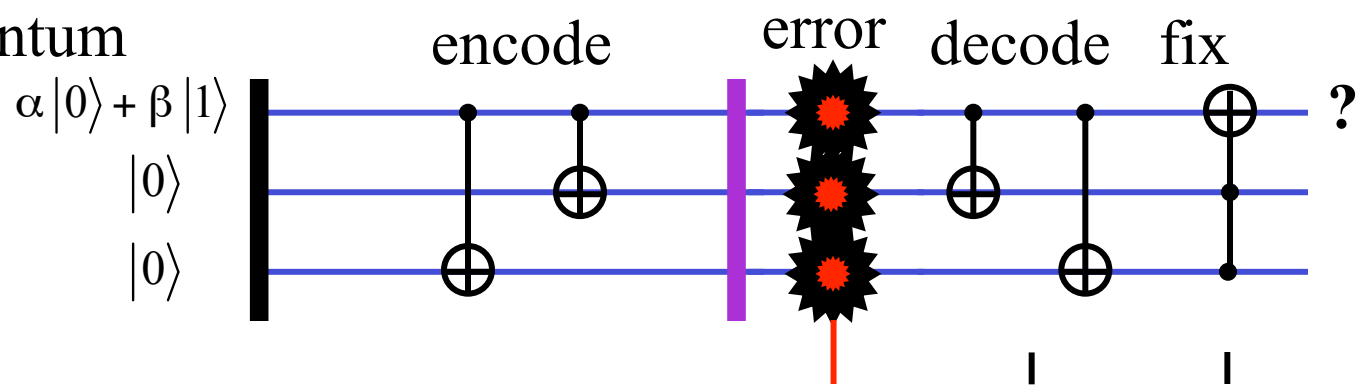


error #@%

Naïve

4. operator identities still hold

Lets be naïve, take classical and move over to quantum



$$(\alpha|0\rangle + \beta|1\rangle) \otimes |0\rangle \otimes |0\rangle \xrightarrow{U} \alpha|0\rangle \otimes |0\rangle \otimes |0\rangle + \beta|1\rangle \otimes |1\rangle \otimes |1\rangle$$

$$\alpha|000\rangle + \beta|111\rangle \overset{XII}{\rightarrow} \alpha|100\rangle + \beta|011\rangle \rightarrow (\alpha|1\rangle + \beta|0\rangle)|11\rangle \rightarrow (\alpha|0\rangle + \beta|1\rangle)|11\rangle$$

$$\alpha|000\rangle + \beta|111\rangle \overset{IXI}{\rightarrow} \alpha|010\rangle + \beta|101\rangle \rightarrow (\alpha|0\rangle + \beta|1\rangle)|10\rangle \rightarrow (\alpha|0\rangle + \beta|1\rangle)|10\rangle$$

$$\alpha|000\rangle + \beta|111\rangle \overset{IIX}{\rightarrow} \alpha|001\rangle + \beta|110\rangle \rightarrow (\alpha|0\rangle + \beta|1\rangle)|01\rangle \rightarrow (\alpha|0\rangle + \beta|1\rangle)|01\rangle$$

error decode fix

2. errors take to orthogonal subspaces + maintain orthogonality

3. syndrome

encoded into subspace:

$|0\rangle \rightarrow |000\rangle, |1\rangle \rightarrow |111\rangle$

(no-cloning evaded!)

$\mathbf{XII}|000\rangle = |100\rangle, \mathbf{XII}|111\rangle = |011\rangle$

OK Wise Guy

What about "phase" errors? $\boldsymbol{\rho}_{out} = (1-p)\boldsymbol{\rho}_{in} + p\boldsymbol{\sigma}_z\boldsymbol{\rho}_{in}\boldsymbol{\sigma}_z$

phase error: $\boldsymbol{\sigma}_z(\alpha|0\rangle + \beta|1\rangle) = (\alpha|0\rangle - \beta|1\rangle)$ …sort of not classical error

Wise guy says "basis change please":

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

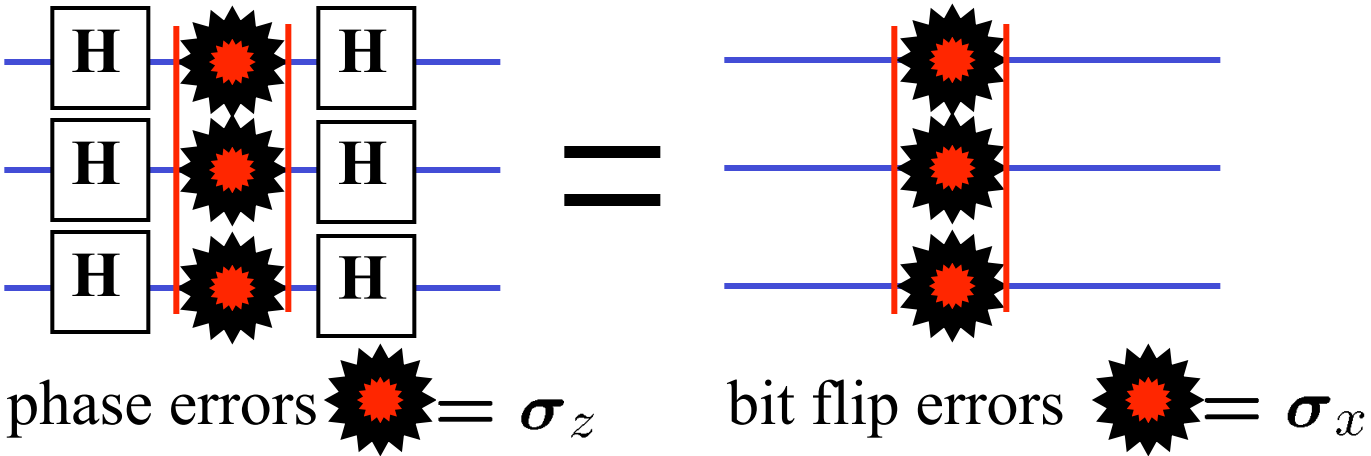
$$|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

$$\boldsymbol{\sigma}_z|+\rangle = |-\rangle$$

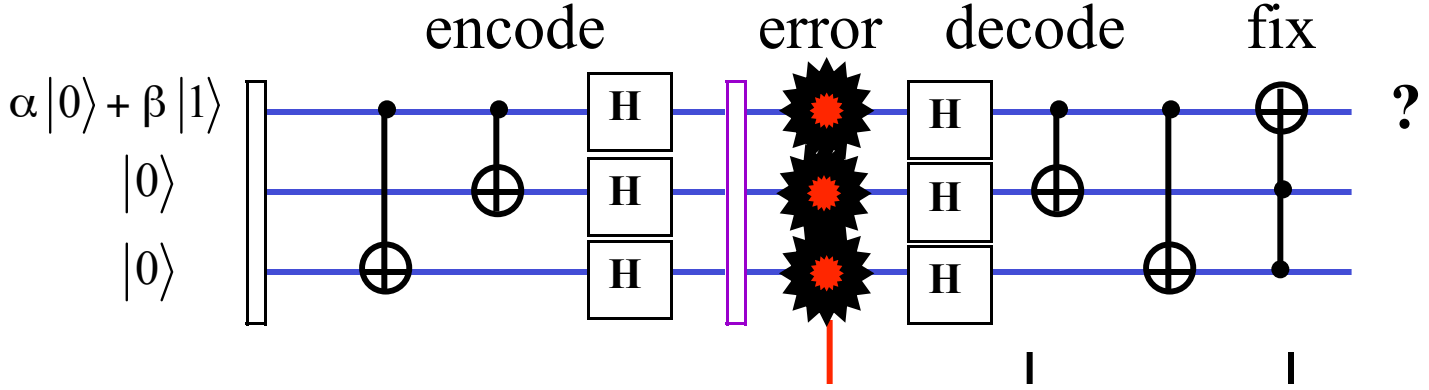
$$\boldsymbol{\sigma}_z|-\rangle = |+\rangle$$

looks like bit flit error in this new basis!

$$\mathbf{H}\boldsymbol{\sigma}_z\mathbf{H} = \boldsymbol{\sigma}_x, \quad \mathbf{H} = |+\rangle\langle 0| + |-\rangle\langle 1|$$



Molly: "I love you, I really love you"
Sam: "Ditto."



$$(\alpha|0\rangle + \beta|1\rangle) \otimes |0\rangle \otimes |0\rangle \xrightarrow{U} \alpha|+\rangle \otimes |+\rangle \otimes |+\rangle + \beta|-\rangle \otimes |-\rangle \otimes |-\rangle$$

$$\alpha|+++\rangle + \beta|---\rangle \overset{ZII}{\rightarrow} \alpha|-++\rangle + \beta|+--\rangle \rightarrow (\alpha|1\rangle + \beta|0\rangle)\,|11\rangle \rightarrow (\alpha|0\rangle + \beta|1\rangle)\,|11\rangle$$

$$\alpha|+++\rangle + \beta|---\rangle \overset{IZI}{\rightarrow} \alpha|+-+\rangle + \beta|-+-\rangle \rightarrow (\alpha|0\rangle + \beta|1\rangle)\,|10\rangle \rightarrow (\alpha|0\rangle + \beta|1\rangle)\,|10\rangle$$

$$\alpha|+++\rangle + \beta|---\rangle \overset{IIZ}{\rightarrow} \alpha|++-\rangle + \beta|--+\rangle \rightarrow (\alpha|0\rangle + \beta|1\rangle)\,|01\rangle \rightarrow (\alpha|0\rangle + \beta|1\rangle)\,|01\rangle$$

error decode fix

encoded into subspace: $|0\rangle \rightarrow |+++\rangle, |1\rangle \rightarrow |---\rangle$

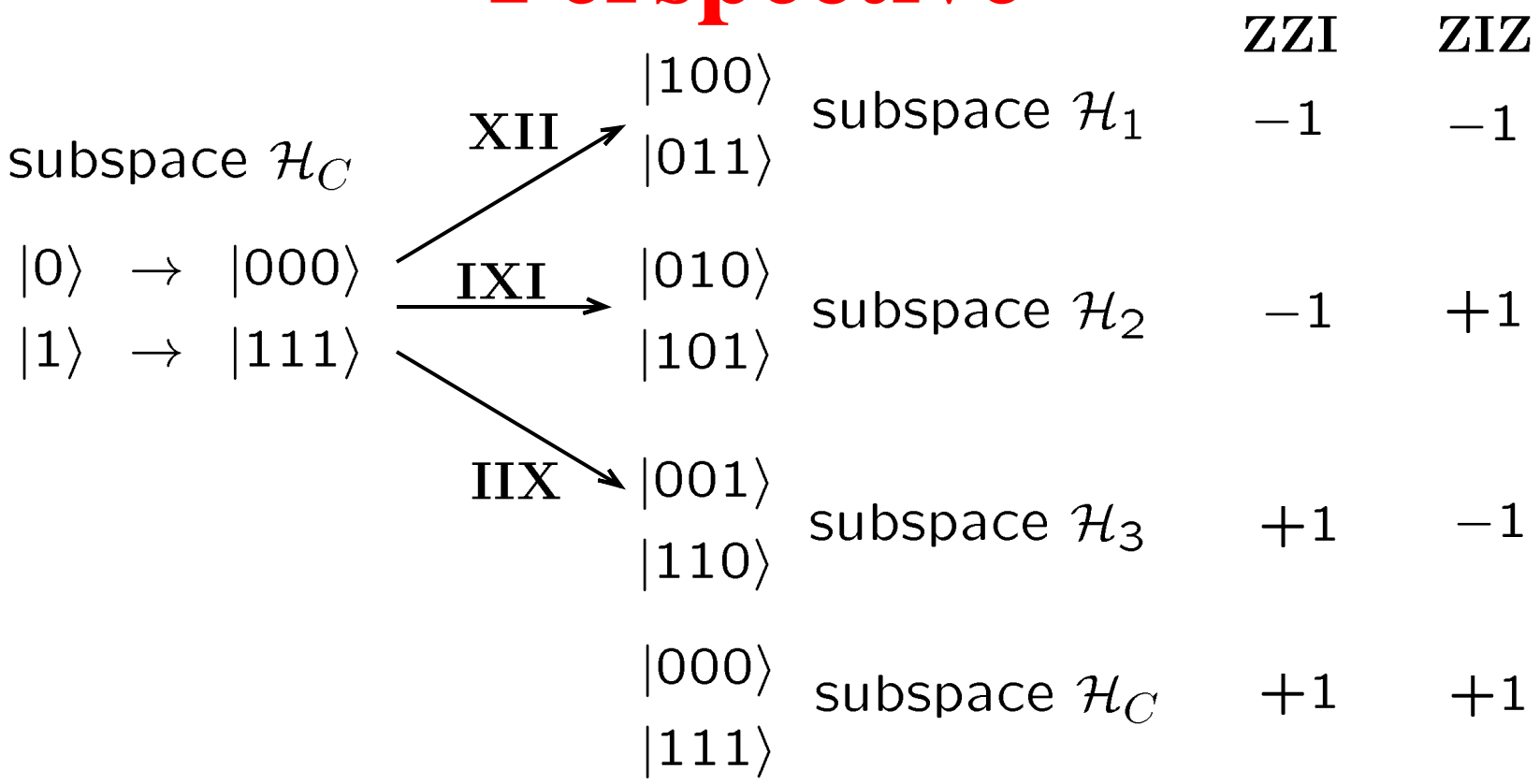
(no-cloning evaded!)

2. errors take to orthogonal subspaces + maintain orthogonality

$\mathbf{ZII}|+++\rangle = |-++\rangle, \mathbf{ZII}|---\rangle = |+--\rangle$

3. syndrome

Perspective



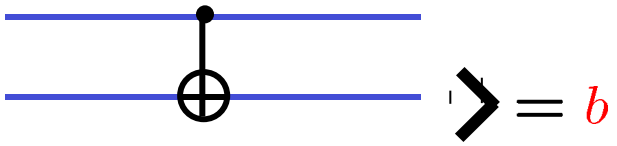
subspaces $\mathcal{H}_C$, $\mathcal{H}_1$, $\mathcal{H}_2$, and $\mathcal{H}_3$ are orthogonal

$\dim \mathcal{H}_C = \dim \mathcal{H}_1 = \dim \mathcal{H}_2 = \dim \mathcal{H}_3 = 2$

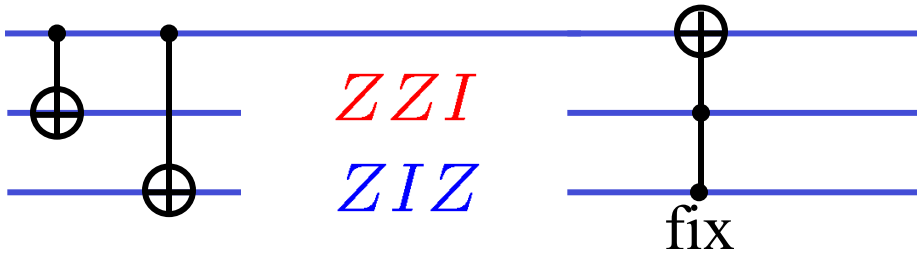
Orthogonal subspaces can be distinguished by measurements

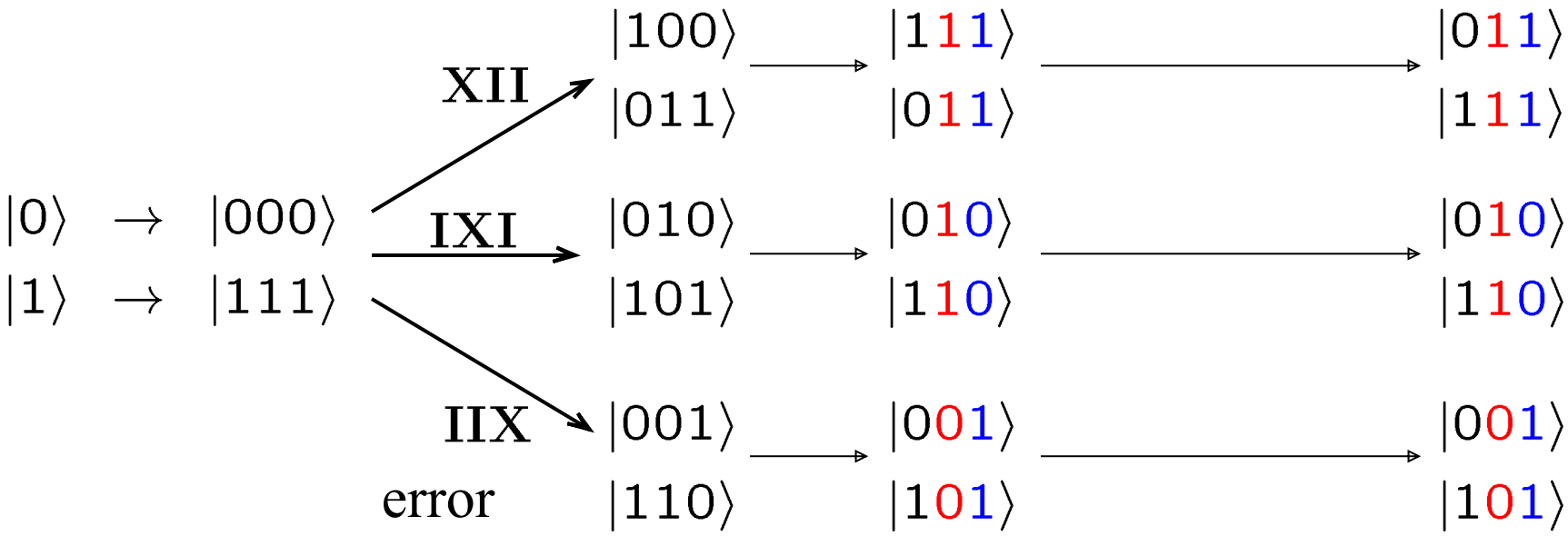
without measuring information encoded into the subspace!!!

Not An Optical Illusion



		b	ZZ
$\|00\rangle$	$\rightarrow$	$\|00\rangle$	$+1$
$\|01\rangle$	$\rightarrow$	$\|01\rangle$	-1
$\|10\rangle$	$\rightarrow$	$\|11\rangle$	-1
$\|11\rangle$	$\rightarrow$	$\|10\rangle$	$+1$





Encoding Away Your Ills

3 qubit bit flip code	3 qubit phase flip code
$\|0\rangle \rightarrow \|000\rangle$	$\|0\rangle \rightarrow \|+++\rangle$
$\|1\rangle \rightarrow \|111\rangle$	$\|1\rangle \rightarrow \|---\rangle$

phase errors **ZII**, **IZI**, **IIZ** act as **Z** on bit flip code qubits:

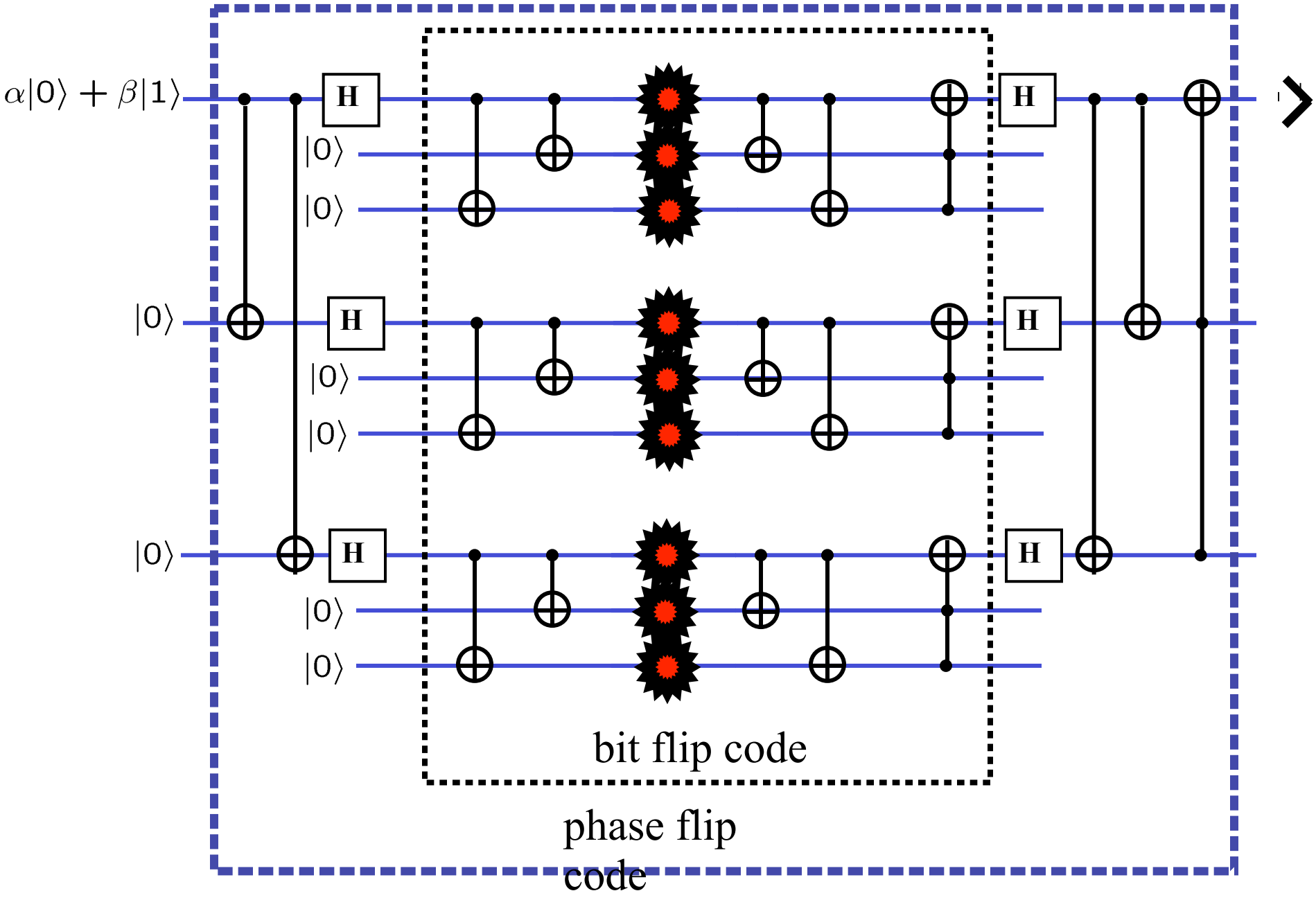
$$|0\rangle_B = |000\rangle \qquad \mathbf{ZII}|0\rangle_B = |0\rangle_B$$
$$|1\rangle_B = |111\rangle \qquad \mathbf{ZII}|1\rangle_B = -|1\rangle_B$$

define: $$|p\rangle = \frac{1}{\sqrt{2}}(|0\rangle_B + |1\rangle_B) = \frac{1}{\sqrt{2}}(|000\rangle + |111\rangle)$$
$$|m\rangle = \frac{1}{\sqrt{2}}(|0\rangle_B - |1\rangle_B) = \frac{1}{\sqrt{2}}(|000\rangle - |111\rangle)$$

Shor Code: (Peter Shor, 1995)

$$|0\rangle \rightarrow |ppp\rangle = \frac{1}{2\sqrt{2}}(|000\rangle + |111\rangle)(|000\rangle + |111\rangle)(|000\rangle + |111\rangle)$$
$$|1\rangle \rightarrow |mmm\rangle = \frac{1}{2\sqrt{2}}(|000\rangle - |111\rangle)(|000\rangle - |111\rangle)(|000\rangle - |111\rangle)$$

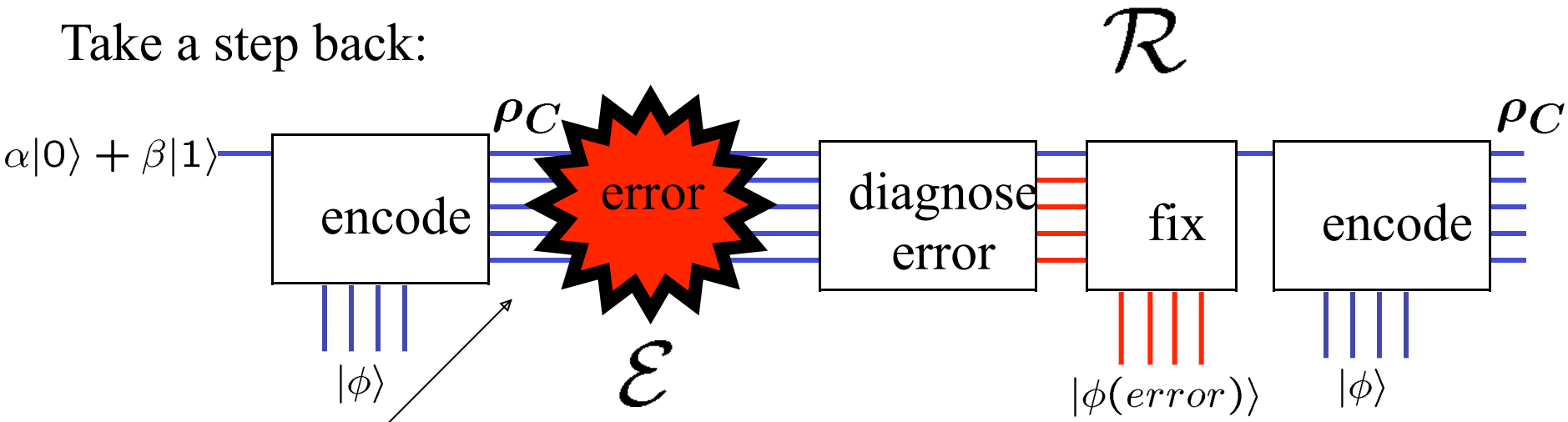
Inside Shor


$\alpha|0\rangle + \beta|1\rangle$
H
$|0\rangle$
$|0\rangle$
$|0\rangle$
H
$|0\rangle$
$|0\rangle$
$|0\rangle$
H
$|0\rangle$
$|0\rangle$
H
H
H
bit flip code
phase flip code

Mysterious Extra Slide

Do the Hokey Pokey

Take a step back:



Here quantum information has simply been encoded into ρ_C

Then an error occurs

$$\mathcal{E}[\rho_C] = \sum_k \mathbf{E}_k \rho_C \mathbf{E}_k^\dagger$$

Then a recovery procedure

$$\mathcal{R}[\mathcal{E}[\rho_C]] = \sum_l \mathbf{R}_l \sum_k \mathbf{E}_k \rho_C \mathbf{E}_k^\dagger \mathbf{R}_l^\dagger$$

An miraculously we recover the encoded state ρ_C

$$\mathcal{R}[\mathcal{E}[\rho_C]] = \rho_C$$

Lets Get Definitional

A quantum code C is simply a ~~subspace~~ subsystem

Quantum Error Correcting Code

A quantum error correcting code C for the quantum operation $\mathcal{E}$ is a code for which there exists a recovery operator $\mathcal{R}$ such that $\mathcal{R} \circ \mathcal{E}$ acts proportional to identity on the codespace.

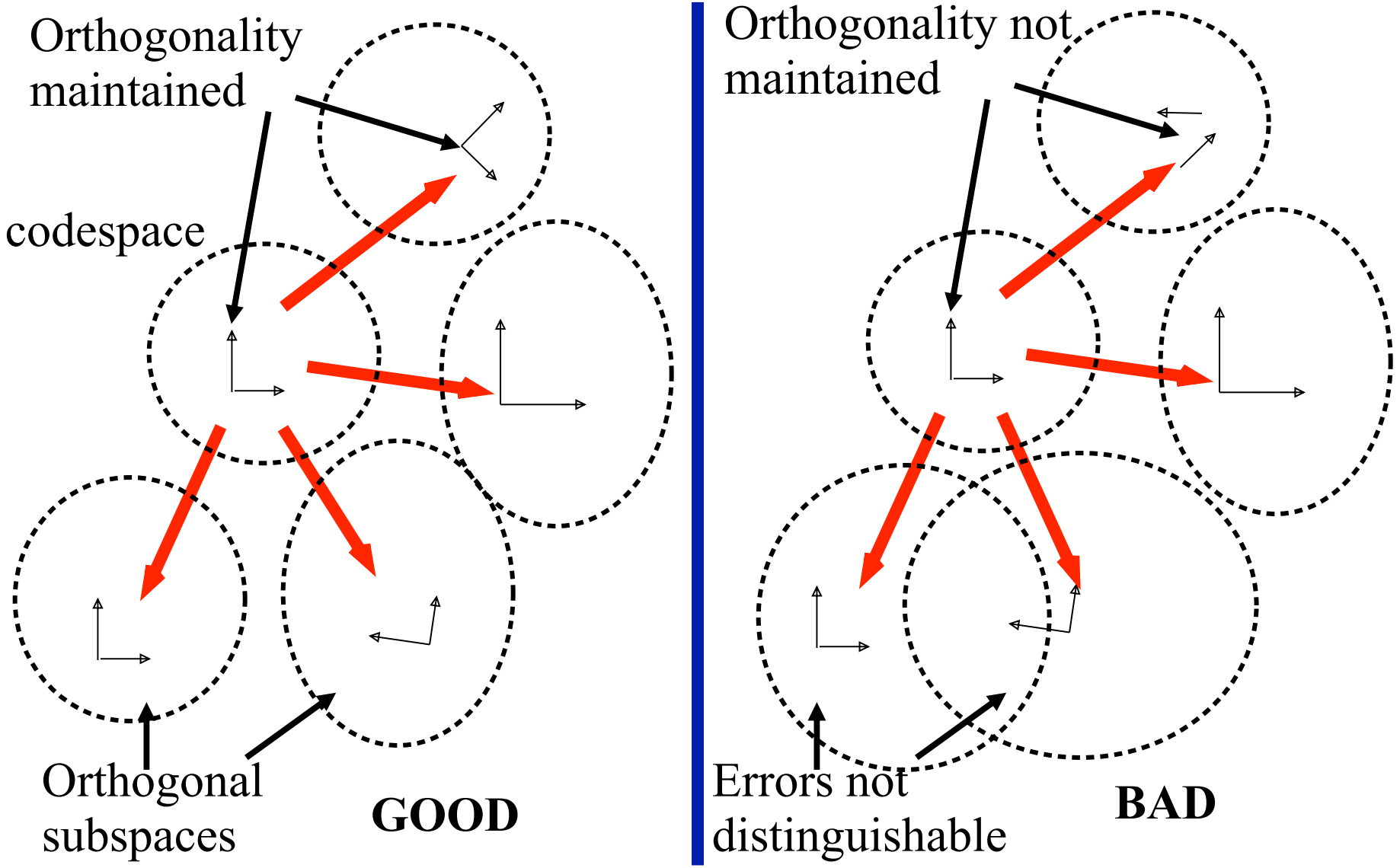
Iff Condition for Quantum Error Correcting Code

A code C with orthonormal basis $\{|i\rangle\}$ is a quantum error correcting code for the quantum operation $\mathcal{E}$ with operation elements $\{\mathbf{E}_k\}$ iff

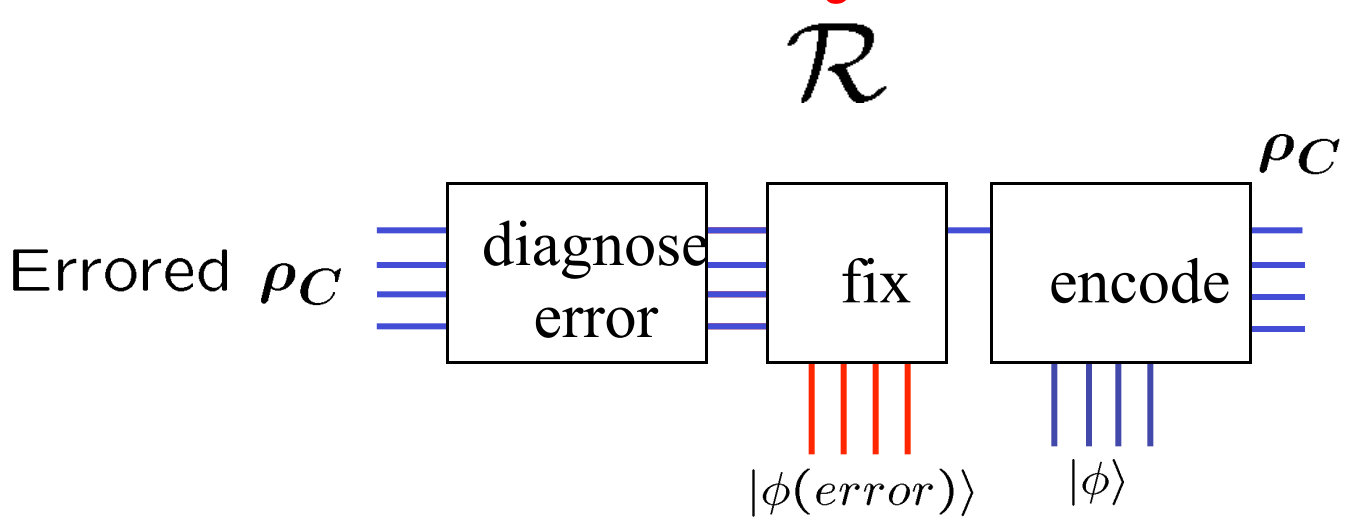
$$\langle i|\mathbf{E}_b^\dagger \mathbf{E}_a|j\rangle = C_{ab}\delta_{ij}$$

Pictures

$$\langle i|\mathbf{E}_a^\dagger \mathbf{E}_b|j\rangle = C_{ab}\delta_{ij}$$



The Recovery Room



Recover $\mathcal{R}$ consists of measurement $\mathcal{P}$ followed by unitary fixing conditional on the measurement outcome:

$$\mathcal{P}[\rho] = \sum_a \mathbf{P}_a \rho \mathbf{P}_a^\dagger$$

$$\mathcal{R}[\rho] = \sum_a \mathbf{U}_a \mathbf{P}_a \rho \mathbf{P}_a \mathbf{U}_a^\dagger$$

$$\boxed{\mathbf{U}_a \mathbf{P}_a \mathbf{E}_b \mathbf{P}_C \propto \mathbf{P}_C}$$

$$\mathbf{P}_C|i\rangle = |i\rangle \text{ iff } |i\rangle \in \text{Code}$$

Digitizing Errors

So far we have designed our error correction for a specific OSR evolution

$$\boldsymbol{\rho} \to \sum_a \mathbf{F}_a \boldsymbol{\rho} \mathbf{F}_a^\dagger$$

such that for our recover operator $\mathcal{R}$

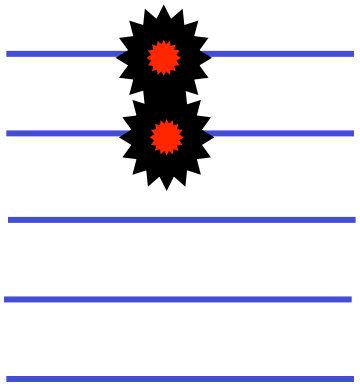
$$\mathcal{R}\left[\sum_a \mathbf{F}_a \boldsymbol{\rho}_C \mathbf{F}_a^\dagger\right] \propto \boldsymbol{\rho}_C$$

If we have another OSR $\rho \to \sum_c \mathbf{E}_c \boldsymbol{\rho} \mathbf{E}_c^\dagger$ whose elements are $\mathbf{E}_c = \sum_a v_{ca} \mathbf{F}_a$, then

$$\mathbf{U}_b \mathbf{P}_b \mathbf{E}_c |i\rangle = \mathbf{U}_b \mathbf{P}_b \sum_a v_{ca} \mathbf{F}_a |i\rangle \propto |i\rangle$$

code corrects OSR operators $\{\mathbf{E}_a\} \to$ code corrects OSR operators $\{\sum_b v_{ba} \mathbf{E}_a\}$.

How Have We Helped?



Suppose error acts on one of many qubits

$$\boldsymbol{\rho} \to \sum_k \mathbf{E}_k \boldsymbol{\rho} \mathbf{E}_k^\dagger, \quad \mathbf{E}_k = \mathbf{E}_k \otimes \mathbf{I} \otimes \cdots \otimes \mathbf{I}$$

Single qubit error correcting code:

$$\text{If } \mathbf{E}_k = e_0 \mathbf{I} + \textstyle\sum_\alpha e_\alpha \boldsymbol{\sigma}_\alpha, \text{ then } \mathcal{R}\left[\sum_k \mathbf{E}_k \boldsymbol{\rho} \mathbf{E}_k^\dagger\right] = \boldsymbol{\rho}$$

$$\mathcal{R}[\boldsymbol{\rho}] = \sum_a \mathbf{U}_a^\dagger \mathbf{P}_a \boldsymbol{\rho} \mathbf{P}_a \mathbf{U}_a$$

Notion of "no error outcome" for a single qubit error, i.e. the measurement projector $\mathbf{P}_0 = \mathbf{P}_C$, implies "probability of no error" $= 1 - p$.

Now suppose two errors act "independently":

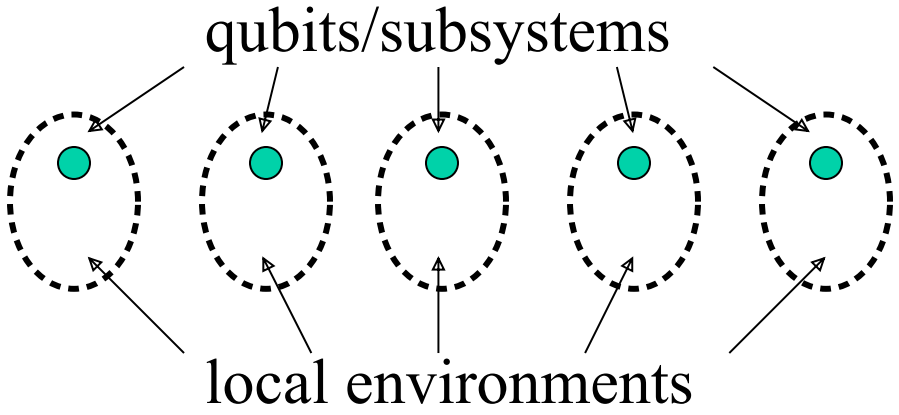
$$\boldsymbol{\rho} \to \sum_l \mathbf{F}_l \sum_k \mathbf{E}_k \boldsymbol{\rho} \mathbf{E}_k^\dagger \mathbf{F}_l^\dagger, \quad \mathbf{E}_k = \mathbf{E}_k \otimes \mathbf{I} \otimes \cdots \otimes \mathbf{I}, \mathbf{F}_l = \mathbf{I} \otimes \mathbf{F}_l \otimes \mathbf{I} \otimes \cdots \otimes \mathbf{I}$$

Notion of no two error probability $= (1-p)^2$

Independent Error Model

?

WARNING: “An approximation only a physicist could love.”



Evolution dominated by interaction with different environment modes which do not interact with each other:

$$\mathcal{E}[\boldsymbol{\rho}] = \sum_{k_1,k_2,\cdots,k_n} \mathbf{E}_{k_1} \otimes \mathbf{E}_{k_2} \otimes \cdots \otimes \mathbf{E}_{k_n} \boldsymbol{\rho} \mathbf{E}_{k_1}^{\dagger} \otimes \mathbf{E}_{k_2}^{\dagger} \otimes \cdots \otimes \mathbf{E}_{k_n}^{\dagger}$$

Notion of “no error probability” gives credence to standard argument for error correction success.

Really we can weaken this requirement (exponentially decreasing correlation [time and space] between errors).

Mysterious Extra Slide II

Linear Classical Codes

Work over binary vector space Z_2^n: $x = (x_1, x_2, \ldots, x_n), x_i \in \{0,1\}$

$x + y = (x_1 \oplus y_1, x_2 \oplus y_2, \ldots, x_n \oplus y_n)$ $\quad < x, y >= \sum_i x_i y_i \mod 2$

k bits encoded into string of length n:

Generator matrix: $G = \begin{pmatrix} v_1 \\ \vdots \\ v_k \end{pmatrix}$ $\quad v_i \in \{0,1\}^n \quad v = \alpha G,\ \alpha \in \{0,1\}^k$

Parity check matrix: $H = \begin{pmatrix} u_1 \\ \vdots \\ u_{n-k} \end{pmatrix}$ $\quad u_i \in \{0,1\}^n \quad Hv = 0$

Effect of error: $w \to w + e$ $\qquad H(w+e) = He =$ syndrome

Set of errors $\{e_i\}$ correctable iff He_i are distinct

Syndrome $He_i \to$ correction is $(w+e_i)+e_i = w$.

Linear Classical Codes II

$[n, k, d]$ code, k bits encoded into n bits.

d = distance of code = $\min_{v \in \mathcal{C}} \text{weight}(v)$

weight(v) = # of 1's in v

distance d code can correct errors of weight t $\quad d = 2t + 1$

Code $\mathcal{C}$: $HG^T = 0$.

Dual code $\mathcal{C}^\perp$: $GH^T = 0$.

$w \in \mathcal{C}^\perp \Leftrightarrow \langle v, w \rangle = 0, v \in \mathcal{C}$

It is possible to have $\mathcal{C}^\perp \subseteq \mathcal{C}$

Example $[n, k, d] = [7, 4, 3]$ code:

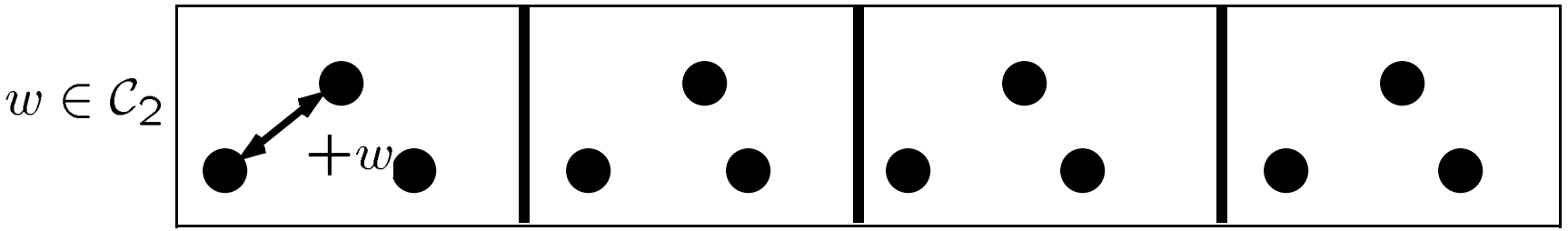
$$H = \begin{pmatrix} 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{pmatrix} \quad G = \begin{pmatrix} 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Dual code is a $[7, 3, 4]$ code

Calderbank-Shor-Steane Codes

code $\mathcal{C}_1$ with subcode $\mathcal{C}_2 \subseteq \mathcal{C}_1$

Equivalence: $u \equiv v \Leftrightarrow \exists w \in \mathcal{C}_2$ s.t. $u = v + w$



Coset of $\mathcal{C}_2$ in $\mathcal{C}_1$ $\qquad$ $\mathcal{C}_1$

CSS quantum code:

$$|\bar{v}\rangle = \frac{1}{\sqrt{2^{k_2}}} \sum_{w \in \mathcal{C}_2} |v + w\rangle$$

$\mathcal{C}_i = [n, k_i, t_i]$ code

$k = k_1 - k_2$ basis elements

$$|\bar{v}\rangle_H = \mathbf{H}^{\otimes n} |\bar{v}\rangle$$

$$\mathbf{H} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

$$|\bar{v}\rangle_H = \frac{1}{\sqrt{2^n}} \sum_{u \in \{0,1\}^n} \frac{1}{\sqrt{2^{k_2}}} \sum_{w \in \mathcal{C}_2} (-1)^{u \cdot (v+w)} |u\rangle$$

$$\sum_{w \in \mathcal{C}} (-1)^{u \cdot w} = \begin{cases} 2^{k_2} \text{ if } u \in \mathcal{C}^{\perp} \\ 0 \text{ otherwise} \end{cases}$$

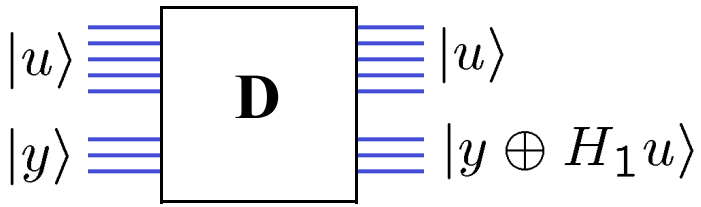
$$|\bar{v}\rangle_H = \frac{1}{\sqrt{2^{n-k_2}}} \sum_{u \in \mathcal{C}_2^{\perp}} (-1)^{u \cdot v} |u\rangle$$

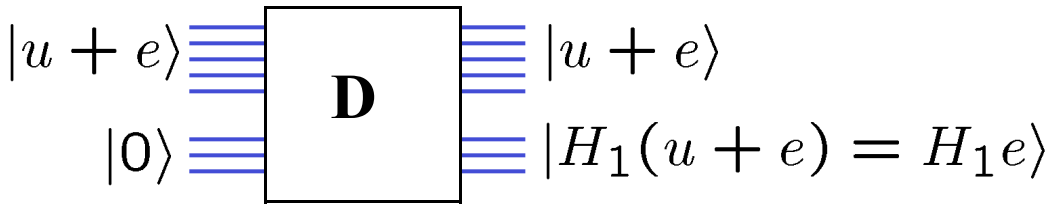
CSS Codes at Work and Play

$$|\bar{v}\rangle = \frac{1}{\sqrt{2^{k_2}}} \sum_{w \in \mathcal{C}_2} |v + w\rangle$$

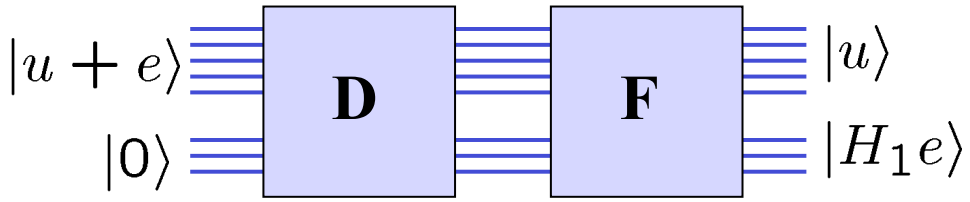
Superposition of $|u\rangle$, $u \in \mathcal{C}_1$.

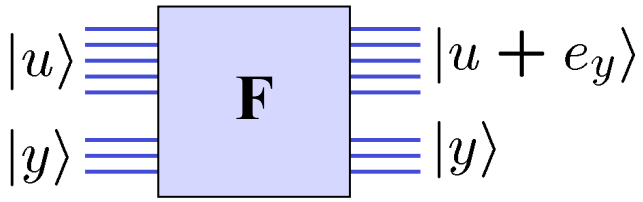
Bit flip error: $\mathbf{X}_e|u\rangle = |u + e\rangle$, $e \in \{0,1\}^n$





H_1=Parity check for $\mathcal{C}_1$





If $\mathcal{C}_1$ is a $[n, k_1, 2t_1 + 1]$ code, we can correct t_1 bit flip errors.

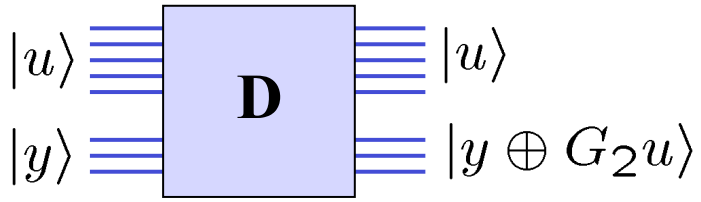
Set Your Phasors To Stun

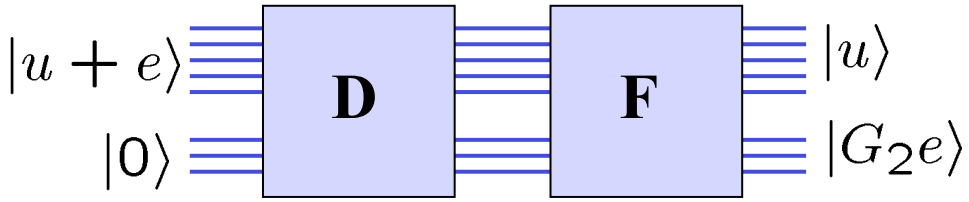
Phase flip error: $\mathbf{Z}_e|u\rangle = (-1)^{u\cdot e}|u\rangle$, $e \in \{0,1\}^n$

$$\mathbf{Z}_e|\bar{v}\rangle_H = \frac{1}{\sqrt{2^{n-k_2}}} \sum_{u\in\mathcal{C}_2^\perp} (-1)^{u\cdot(v+e)}|u\rangle = |\overline{v+e}\rangle_H$$

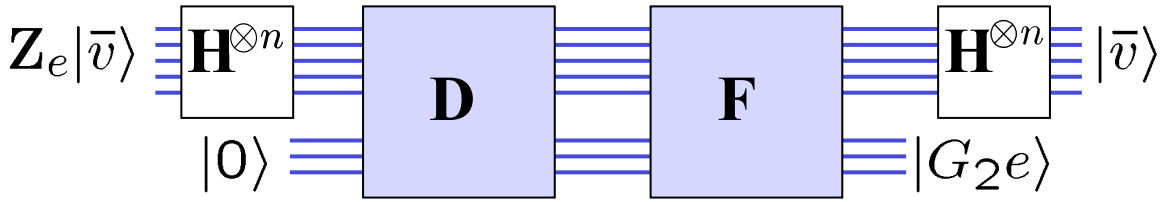
Phase errors look like bit flips in the H basis!

In the H basis, codewords are superpositions of $|u\rangle$, $u \in \mathcal{C}_2$.





G_2=Parity check for $\mathcal{C}_2^\perp$



If $\mathcal{C}_2^\perp$ is a $[n, k_2^\perp, 2t_2^\perp + 1]$ code, we can correct $t_2^\perp$ phase flip errors.

Examplicious

Example $\mathcal{C}_1 = [n, k_1, d_1] = [7, 4, 3]$ code:

$$H_1 = \begin{pmatrix} 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{pmatrix} \qquad G_2^{\perp} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{pmatrix}$$

$\mathcal{C}_2^{\perp} = [n, k_1^{\perp}, d_1^{\perp}] = [7, 3, 4]$ code.

Expliciticious: $[[7, 1, 3]]$ quantum code

$$\begin{aligned} |0_L\rangle &= \frac{1}{\sqrt{8}}(|0000000\rangle + |1010101\rangle + |0110011\rangle + |1100110\rangle \\ &\quad + |0001111\rangle + |1011010\rangle + |0111100\rangle + |1101001\rangle) \\ |1_L\rangle &= \frac{1}{\sqrt{8}}(|1111111\rangle + |01011010\rangle + |1001100\rangle + |0011001\rangle \\ &\quad + |1110000\rangle + |0100101\rangle + |1000011\rangle + |0010110\rangle) \end{aligned}$$

For Those Who Just Woke Up

Real quantum systems are open systems.

so we need

Quantum error correcting codes.

encode into code subspace

digitizing errors

independent error model

quantum error correcting code condition

$$\langle i|\mathbf{E}_b^\dagger \mathbf{E}_a|j\rangle = C_{ab}\delta_{ij}$$

CSS codes

classical linear codes $\rightarrow$ quantum codes



Unmixing Simplification

$\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B$



Purification of system B

$\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B \otimes \mathcal{H}_C$



$\boldsymbol{\rho}_{out} = \sum_k \mathbf{A}_k \boldsymbol{\rho}_A \mathbf{A}_k^\dagger$ where $\mathbf{A}_{k=(\mu,\nu)} = \sqrt{p_\nu}\langle\mu|\mathbf{U}|\nu\rangle$

Purification of system B

$\boldsymbol{\rho}_{out} = \sum_k \mathbf{A}_k \boldsymbol{\rho}_A \mathbf{A}_k^\dagger$ where $\mathbf{A}_k = \langle k|\mathbf{U}|\nu\rangle$

Necessary

$\mathcal{R}[\mathcal{E}[\boldsymbol{\rho}_C]] \propto \boldsymbol{\rho}_C$ for $\boldsymbol{\rho}_C$ with support over the code C:

$$\sum_{kl} \mathbf{R}_l \mathbf{E}_k \boldsymbol{\rho}_c \mathbf{E}_k^\dagger \mathbf{R}_l^\dagger = c\boldsymbol{\rho}_c$$

$\Downarrow$

$\sum_{kl} \mathbf{R}_l \mathbf{E}_k \mathbf{P}_C \boldsymbol{\rho} \mathbf{P}_C \mathbf{E}_k^\dagger \mathbf{R}_l^\dagger = c\mathbf{P}_C \boldsymbol{\rho} \mathbf{P}_C$ where $\mathbf{P}_C = \sum_{i \in C} |i\rangle\langle i|$ is projector onto code C.

$\Downarrow$

This must hold for all $\boldsymbol{\rho}$, so $\mathbf{R}_l \mathbf{E}_k \mathbf{P}_C = \lambda_{lk} \mathbf{P}_C$

$\Downarrow$

$$\mathbf{P}_C \mathbf{E}_m^\dagger \mathbf{R}_l^\dagger \mathbf{R}_l \mathbf{E}_k \mathbf{P}_C = \lambda_{lm}^* \lambda_{lk} \mathbf{P}_C$$

$\Downarrow$

Use $\sum_l \mathbf{R}_l^\dagger \mathbf{R}_l = \mathbf{I}$, to obtain $\mathbf{P}_C \mathbf{E}_m^\dagger \mathbf{E}_k \mathbf{P}_C = \sum_l \lambda_{lm}^* \lambda_{lk} \mathbf{P}_C$

$\Downarrow$

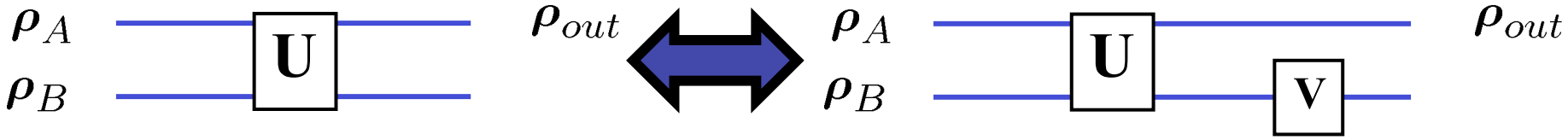
$$\langle i | \mathbf{E}_m^\dagger \mathbf{E}_k | j \rangle = C_{mk} \delta_{ij}$$

OSR Unitary Freedom

OSR: $\mathbf{A}_k = \langle k|\mathbf{U}|\nu\rangle$ $\quad$ $\boldsymbol{\rho}_{out} = \sum_k \mathbf{A}_k \boldsymbol{\rho}_{in} \mathbf{A}_k^\dagger$

Whose $|k\rangle$?

$|k\rangle$ was from Trace operation: basis arbitrary.



Equivalent from system A's perspective

$$|l\rangle = \sum_k V_{kl}|k\rangle, |k\rangle = \sum_l V_{kl}^*|l\rangle, \quad VV^\dagger = I$$

New OSR: $\mathbf{B}_l = \langle l|\mathbf{U}|\nu\rangle$

$$\mathbf{A}_k = \langle k|\mathbf{U}|\nu\rangle = \sum_l V_{kl}\langle l|\mathbf{U}|\nu\rangle = \sum_l V_{kl}\mathbf{B}_l$$

Two OSR's $\{\mathbf{A}_k\}$ and $\{\mathbf{B}_k\}$ are equivalent iff there exists a unitary matrix V_{kl} such that $\mathbf{A}_k = \sum_l V_{kl}\mathbf{B}_l$.

Diagonal Error Condition

Error correcting code criteria: $\langle i|\mathbf{E}_a^\dagger \mathbf{E}_b|j\rangle = C_{ab}\delta_{ij}$

$$C_{ab}\delta_{ij} = \langle i|\mathbf{E}_a^\dagger \mathbf{E}_b|j\rangle = \left(\langle j|\mathbf{E}_b^\dagger \mathbf{E}_a|i\rangle\right)^* = C_{ba}^*\delta_{ij}$$

$C_{ab} = C_{ba}^*$: C is Hermitian

Change of OSR basis: $\mathbf{F}_k = \sum_k u_{ka}\mathbf{E}_a$

Error correcting code criteria: $\langle i|\mathbf{F}_k^\dagger \mathbf{F}_l|j\rangle = D_{kl}\delta_{ij}$

$$\langle i|\mathbf{F}_k^\dagger \mathbf{F}_l|j\rangle = \sum_{ab} u_{ka}^* u_{lb}\langle i|\mathbf{E}_a^\dagger \mathbf{E}_b|j\rangle = \sum_{ab} u_{ka}^* C_{ab} u_{bl}\delta_{ij}$$

$$\sum_{ab} u_{ka}^* C_{ab} u_{bl} = D_{kl} = d_k\delta_{kl}$$

$$\boxed{\langle i|\mathbf{F}_a^\dagger \mathbf{F}_b|j\rangle = d_a\delta_{ab}\delta_{ij}}$$

Sufficient

Projector onto the code: $\mathbf{P}_C = \sum_{i \in C} |i\rangle\langle i|$

$$\langle i|\mathbf{F}_a^\dagger \mathbf{F}_b|j\rangle = d_a \delta_{ab}\delta_{ij}$$

Projector onto the code followed by error $\mathbf{F}_a$:

$$\mathbf{F}_a\mathbf{P}_C = \mathbf{U}_a\sqrt{\mathbf{P}_C\mathbf{F}_a^\dagger\mathbf{F}_a\mathbf{P}_C} = \mathbf{U}_a\sqrt{d_a}\mathbf{P}_C$$

Polar decomposition: $\mathbf{A} = \mathbf{U}\sqrt{\mathbf{A}^\dagger\mathbf{A}}$, $\mathbf{U}$ unitary

Define the projectors $\mathbf{P}_a = \mathbf{U}_a\mathbf{P}_C\mathbf{U}_a^\dagger$ which are orthogonal:

$$\mathbf{P}_a\mathbf{P}_b = \mathbf{U}_a\mathbf{P}_C\mathbf{U}_a^\dagger\mathbf{U}_b\mathbf{P}_C\mathbf{U}_b^\dagger = \frac{1}{\sqrt{d_a d_b}}\mathbf{U}_a\mathbf{P}_C\mathbf{F}_a^\dagger\mathbf{F}_b\mathbf{P}_C\mathbf{U}_b^\dagger = \delta_{ab}$$

$$\mathbf{U}_a^\dagger\mathbf{P}_a\mathbf{F}_a\mathbf{P}_C = \sqrt{d_a}\mathbf{P}_C$$

fix/diagnose error/error/code

$$\mathbf{U}_b^\dagger\mathbf{P}_b\mathbf{F}_a\mathbf{P}_C = 0$$

fix/diagnose no error/error/code

$$\mathbf{U}_a^\dagger \mathbf{P}_a \mathbf{F}_a \mathbf{P}_C = \sqrt{d_a}\mathbf{P}_C \qquad \mathbf{U}_b^\dagger \mathbf{P}_b \mathbf{F}_a \mathbf{P}_C = 0$$

Recover $\mathcal{R}$ consists of measurement $\mathcal{P}$ followed by unitary fixing conditional on the measurement outcome:

Padding may be needed.

$$\mathcal{P}[\boldsymbol{\rho}] = \sum_a \mathbf{P}_a \boldsymbol{\rho} \mathbf{P}_a^\dagger$$

$$\mathcal{R}[\boldsymbol{\rho}] = \sum_a \mathbf{U}_a^\dagger \mathbf{P}_a \boldsymbol{\rho} \mathbf{P}_a \mathbf{U}_a$$

$$\mathcal{R} \circ \mathcal{F}[\boldsymbol{\rho}_C] = \textstyle\sum_{ab} \mathbf{U}_a^\dagger \mathbf{P}_a \mathbf{F}_b \mathbf{P}_C \boldsymbol{\rho}_C \mathbf{P}_C \mathbf{F}_b^\dagger \mathbf{P}_a \mathbf{U}_a \propto \boldsymbol{\rho}_C$$

recovery error

Explicit recovery operator derived.